

AARON RHIM

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Email | [LinkedIn](#) | [GitHub](#) | [Portfolio Website](#)

EDUCATION

University of British Columbia (UBC)

Vancouver, BC

Combined Major in Computer Science, Mathematics, and Physics

Sep 2024 – Apr 2028

- **Relevant Coursework:** Machine Learning, Data Structures & Algorithms (Intro & Intermediate), Computer Systems, Computer Hardware and Operating Systems, Linear Algebra (Honors), Discrete Mathematics

RELEVANT EXPERIENCE

Founder – Eithelmir

May 2026 – Present

Python, Deep Learning, Machine Learning, Sensor Fusion, Web design

Vancouver, British Columbia

- Built a multi-modal pipeline that **predicts half-body pose from egocentric video capture and two wrist IMUs**, reaching 9.8 cm mean joint error—validated using multi-view RGB triangulation
- Designed a development interface that allows **visualization of 3D limb localization and reconstruction** using a combination of SMPL avatars, MANO hand rigs, and controlled recording sessions
- Developing a website that handles payment traffic and cloud-based data infrastructure: <https://app.eithelmir.com>

Software Co-Lead – UBC Rover

Sep. 2025 – Present

C++, ROS2, MoveIt2, RViz2, CAN-FD, Python, Moteus, Git, MuJoCo, Robosuite

Vancouver, British Columbia

- Built two **human-machine interfaces** (HMI) for intuitive debugging for non-software members and another one for the operator to use during competition
- Designed a **reinforcement learning** pipeline for the team's custom 6-DoF robotic arm, reaching **98%** task completion on a keyboard-typing task
- Programmed software to **improve** source code modularity, kinematic arm control, development operations, and existing serial communication methods
- Created software **solutions** for GNSS mapping and reliable digital twin visualization
- Debugged electrical and mechanical assembly of the custom 6-DOF robotic arm under competition stress, **successfully pinpointing concentric error** in a high-load-bearing motor and helping design a solution using CAD
- Designed the team website: <https://www.ubcrover.com>

PROJECT PORTFOLIO

3D Rendering Engine + ANN Visualization | Java, Multithreading

[Github Link](#)

- Built a **3D rendering engine** from scratch in Java
- Implemented core **linear algebra** operations for object transformations, projection, camera movement, and interactive scene navigation
- Integrated **artificial neural network visualization** modules, including **gradient-descent demos**, to make model behavior and learning dynamics easier to inspect interactively

An actually good job finder | Web development

[Github Link](#)

- Built a platform that finds and notifies me instantly of internship listings for saved companies
- Builds a prospective listings calendar based on retrospective company listings using RNNs

AWARDS

- **Top 10% Y-Combinator** applicants (F26)
- Hackathon Awards: **CS6 Hacks** (Best Solo Hacker), **Kickstart** (3rd-place), **StormHacks** (Best Hardware)

TECHNICAL SKILLS

Programming Languages: Python, Java, C++/C, JavaScript, HTML/CSS, Assembly (Y86)

Frameworks & Libraries: RoboSuite, Flask, React, Node.js, TensorFlow, PyTorch, Tailwind, PostgreSQL

Tools & Platforms: Git, Docker, ROS2, CAN, Moteus, MoveIt2, RViz2, AWS (EC2, S3, DynamoDB, Bedrock), Linux (Arch, Ubuntu), Supabase, Framer Motion